Introduction to Probability

How to think clearly about chance

Part A · Chance and counting

why probability, sets, counting, basic rules

Part B · Updating with information

conditional probability, Bayes' theorem

Part C · Random variables

random variables, distributions, two variables

Ibrahim Sobh, PhD · ibrahimsobh.github.io/introduction-to-probability

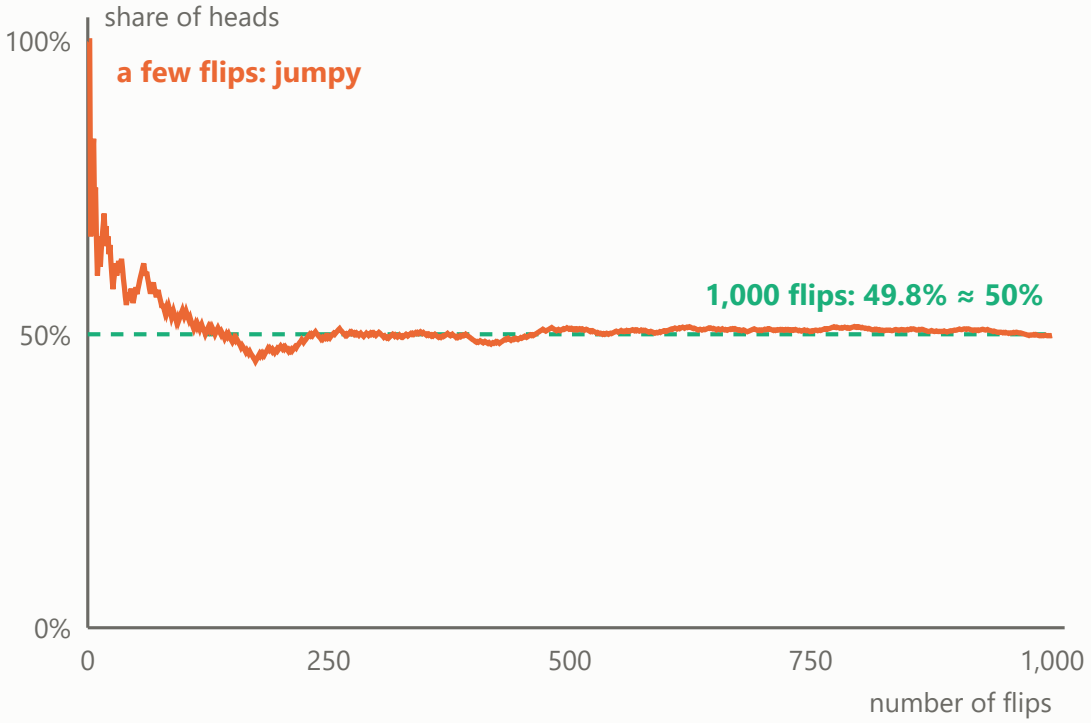
PART A

Chance and counting

1 Why probability?
2 Sets and sample spaces
3 Counting
4 Basic probability rules

Many trials reveal the true probability

- A **probability** is a number from 0 (can't happen) to 1 (certain), often written as a percentage
- Three ways to find one: **classical** (count equally likely outcomes: P(heads) = ½), **frequency** (the bus was late 6 of 50 days: ≈ 12%), **subjective** (a reasoned degree of belief)
- Experiment → outcome → event: rolling a die → a 4 → "an even number"



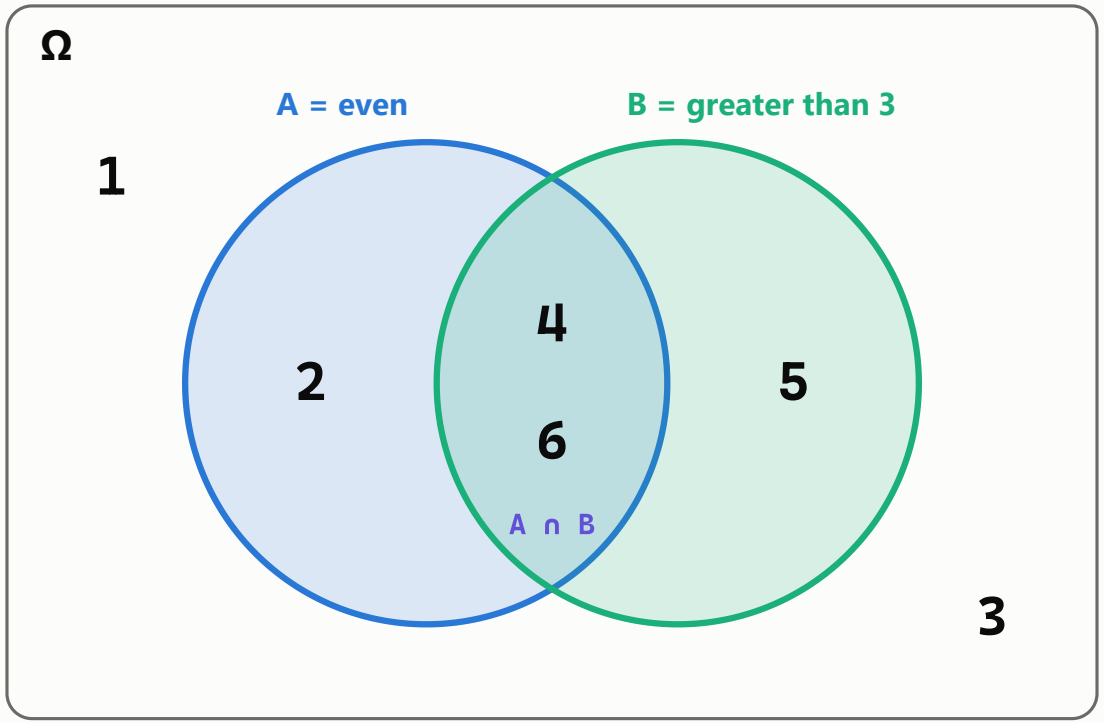
KEY INSIGHT

With a few trials the share can be far from the probability; with many, it settles close and stays close (the law of large numbers).

Events are sets of outcomes

- The **sample space Ω** is the set of all outcomes: a die roll gives Ω = {1, 2, 3, 4, 5, 6}
- An **event** is a part of Ω: "even" = {2, 4, 6}

Operation	Symbol	Die (A = even, B = greater than 3)
Union: A or B	A ∪ B	{2, 4, 5, 6}
Intersection: A and B	A ∩ B	{4, 6}
Complement: not A	A^c	{1, 3, 5}

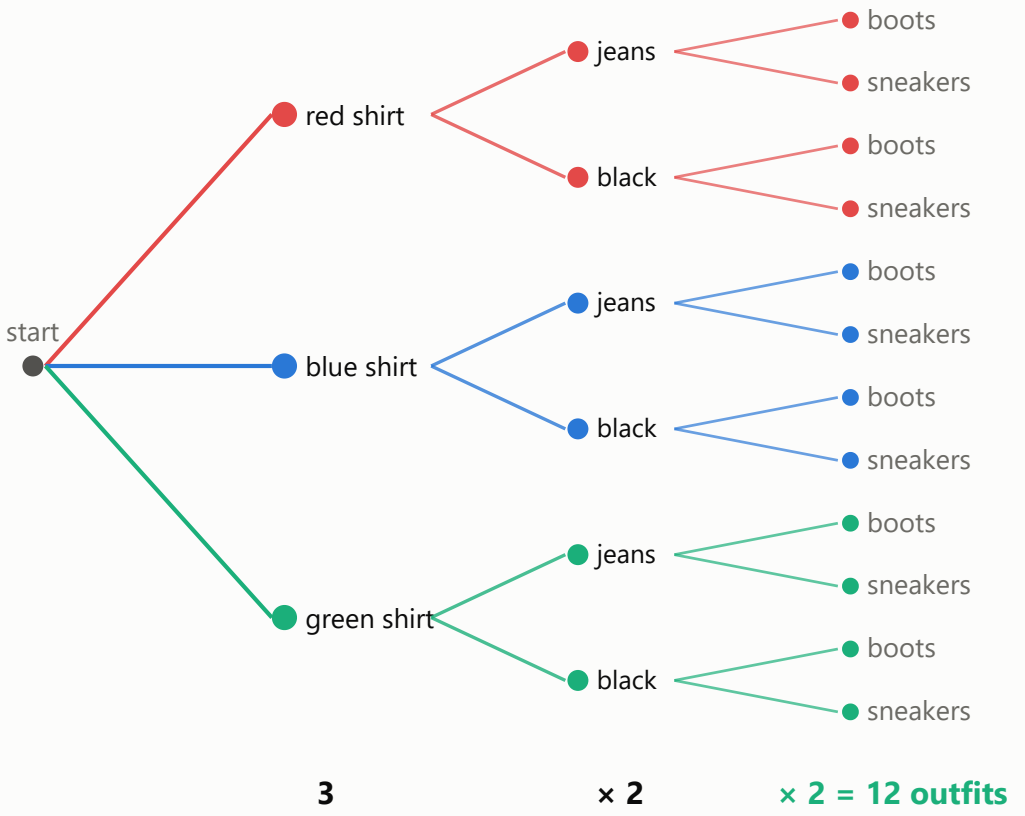


KEY INSIGHT

Every probability question starts the same way: what is Ω, and which part of it is my event?

Choices in steps multiply

- **Multiplication rule:** a choices, then b, then c → a × b × c results
- 3 shirts × 2 trousers × 2 pairs of shoes = **12 outfits**
- A 4-digit PIN: 10 × 10 × 10 × 10 = **10,000**
- 8 small letters: 26^8 ≈ 209 billion; with capitals and digits (62 choices): ≈ 218 trillion



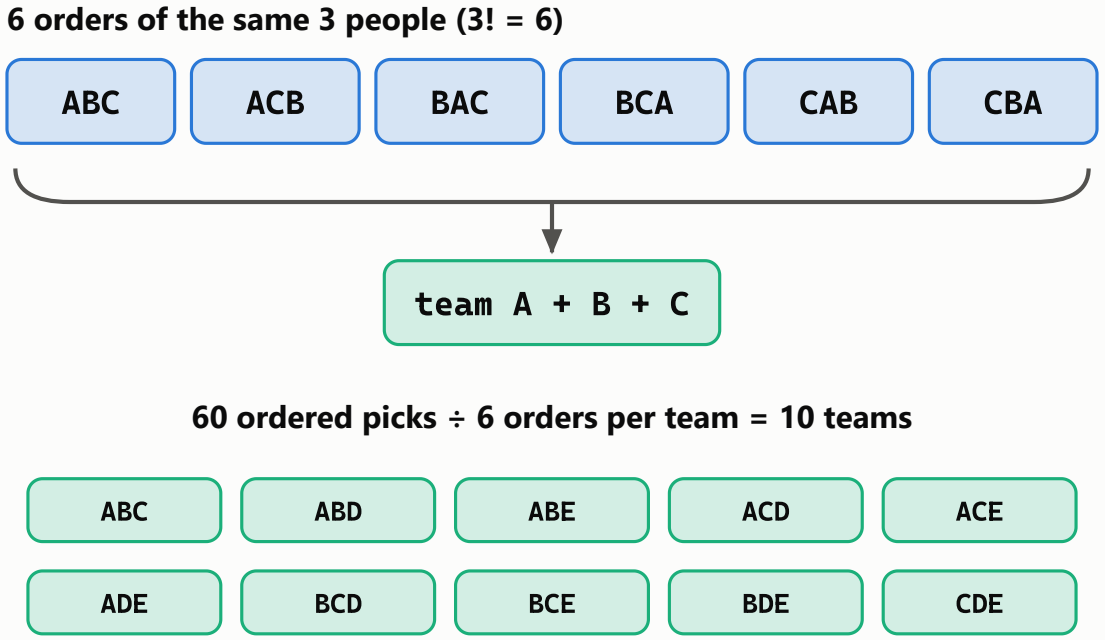
KEY INSIGHT

Each extra step multiplies the total by its number of choices.

Does order matter? Permutations vs combinations

	Permutations	Combinations
Order matters?	yes: ABC ≠ CBA	no: ABC = CBA
Choose 3 of 5	5 × 4 × 3 = **60**	60 ÷ 3! = **10**
Examples	medals, a running order	a team, pizza toppings

$n! = n \times (n-1) \times \ldots \times 1$ counts the ways to put n things in order: 3! = 6



KEY INSIGHT

If swapping two choices gives a different result, order matters. If not, count the ordered picks and divide by k!.

If events overlap, subtract the overlap once

- Equally likely outcomes: **P(A) = favourable ÷ total**
- **Addition rule: P(A ∪ B) = P(A) + P(B) − P(A ∩ B)**
- P(heart or face card) = 13/52 + 12/52 − 3/52 = **22/52 ≈ 0.42**
- If A and B can't happen together, the overlap is 0: just add



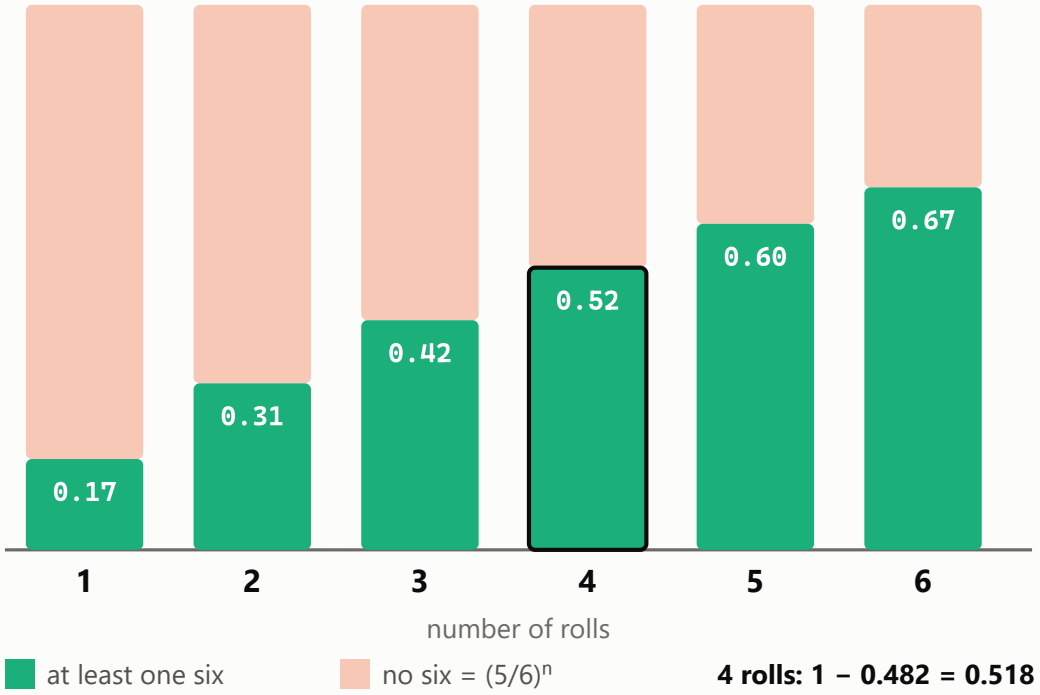

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P(heart or face) = 13/52 + 12/52 − 3/52 = 22/52 ≈ 0.42
```

KEY INSIGHT

Adding two events counts their overlap twice, so subtract it once.

"At least one" is easier as "1 − none"

- **Complement rule: $P(A^c) = 1 - P(A)$**: P(not a six) = 1 − 1/6 = 5/6
- Roll a die 4 times: P(no six) = $(5/6)^4$ = 625 ÷ 1,296 ≈ 0.482
- P(at least one six) = 1 − 0.482 = **0.518**, a little more than half



KEY INSIGHT

When a question says "at least one", compute 1 − P(none) first.

PART B

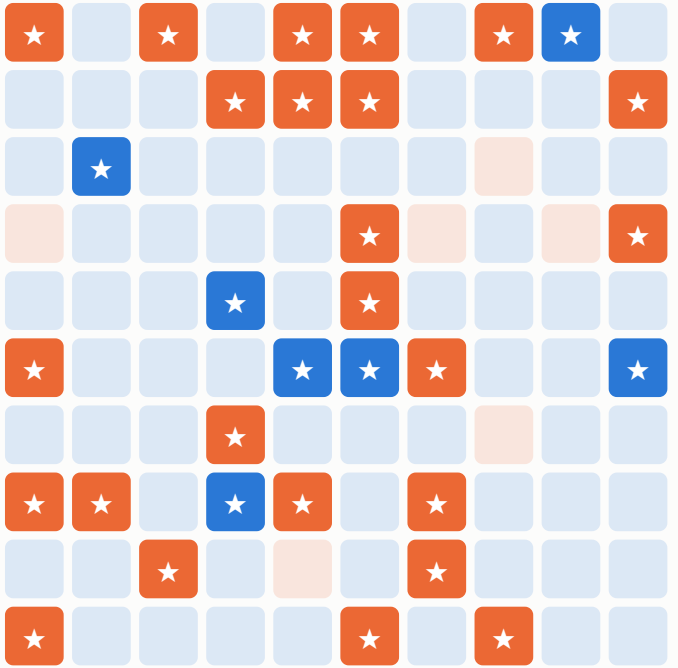
Updating with information

5 Conditional probability

6 Bayes' theorem

Conditioning means zooming in

- **P(A | B)** = the chance of A once we know B happened: only outcomes in B count now
- **`P(A | B) = P(A ∩ B) ÷ P(B)`**: die, A = even, B = greater than 3: (2/6) ÷ (3/6) = 2/3
- 100 emails: 30 spam, 31 contain "free", 24 both → P(spam | free) = 24/31 ≈ **0.77**, but P(free | spam) = 24/30 = **0.80**



Zoom into the 31 emails that contain "free" ★

24 of them are spam:

`P(spam | free)`
`= 24/31 ≈ 0.77`

Before zooming:

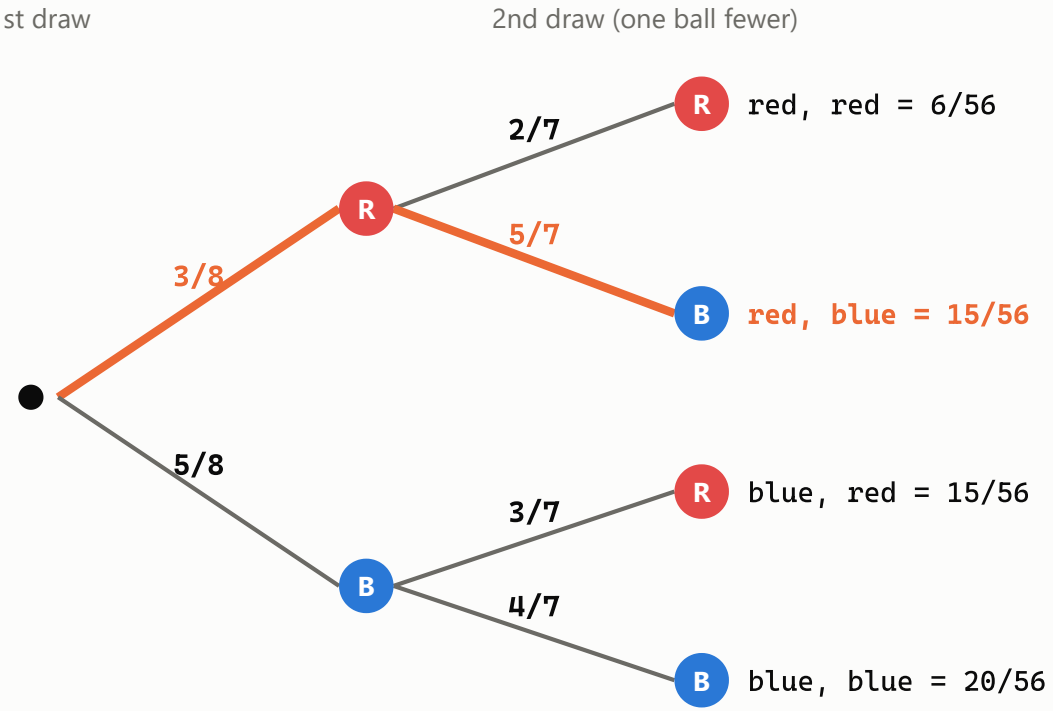
`P(spam) = 30/100 = 0.30`

KEY INSIGHT

Knowing B changes which group you look at. And P(A | B) is not P(B | A).

Multiply along the path; independent events don't care

- **Multiplication rule: `P(A and B) = P(A) × P(B | A)`**
- A box of 3 red and 5 blue, two draws without putting back: P(red, then blue) = 3/8 × 5/7 = **15/56**
- **Independent:** knowing A doesn't change B, P(B | A) = P(B), so P(A and B) = P(A) × P(B): heads and an even die = ½ × ½ = ¼

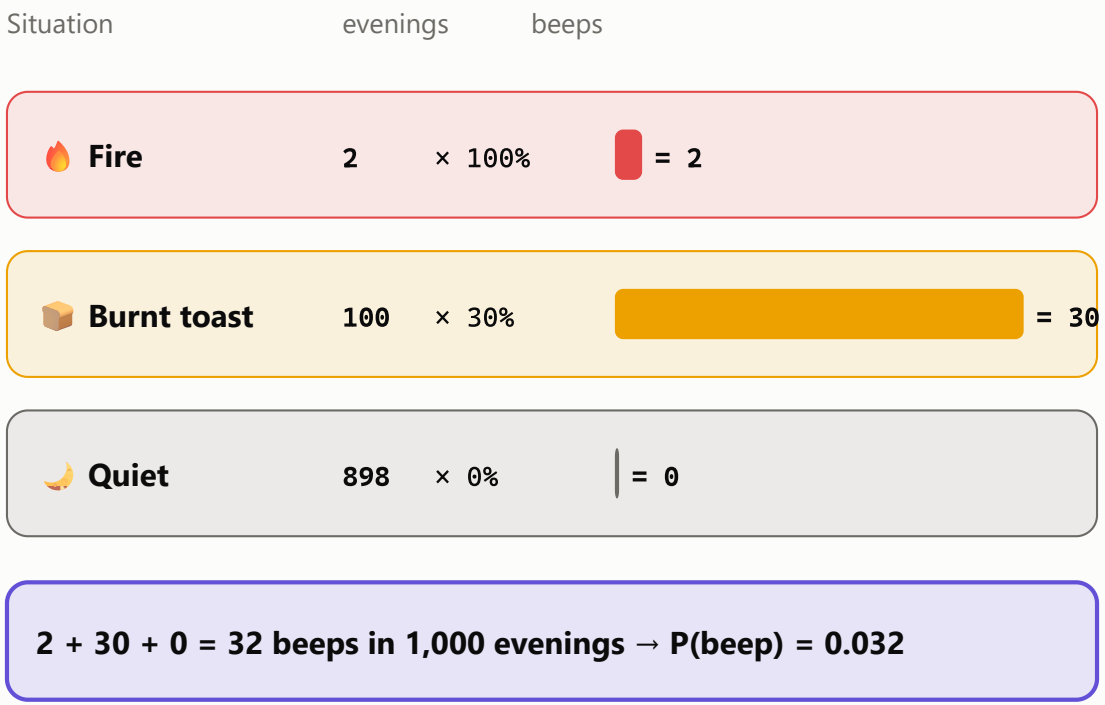


KEY INSIGHT

A coin has no memory: after five heads, the next flip is still 50/50 (the gambler's fallacy).

Add up every way the evidence can happen

- **Law of total probability: $P(B) = \Sigma_i\ P(B \mid A_i) \times P(A_i)$**, over slices A_i that don't overlap and cover everything
- A smoke alarm over 1,000 evenings: a fire on 2 (beeps every time) → 2 beeps; burnt toast on 100 (beeps 30% of the time) → 30 beeps; 898 quiet evenings → 0
- P(beep) = 0.002 × 1 + 0.10 × 0.30 = **0.032**

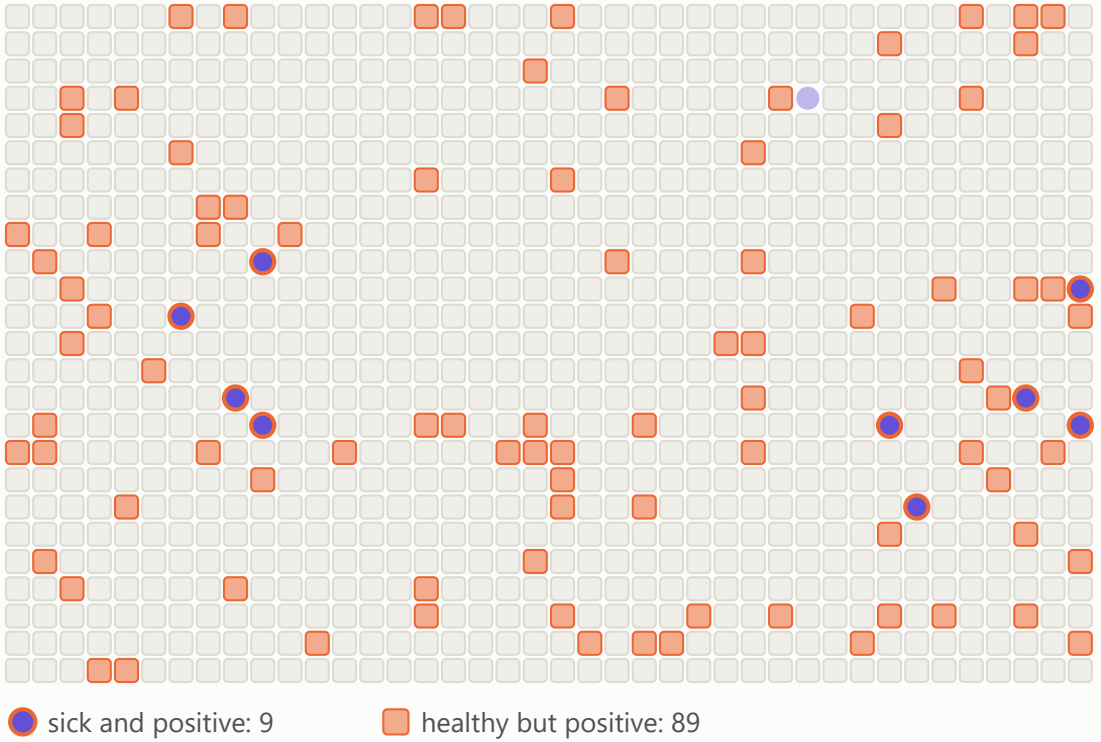


KEY INSIGHT

Split the world into separate situations, find the event's chance in each, and add.

Bayes: always ask for the base rate

- **`P(A | B) = P(B | A) × P(A) ÷ P(B)`**: from P(evidence | cause) to P(cause | evidence)
- Smoke alarm: P(fire | beep) = 1 × 0.002 ÷ 0.032 ≈ **6%** (2 of the 32 beeps)
- Medical test on 1,000 people: 1% sick (10), test finds 90% (9), 9% false alarms on the 990 healthy (89) → P(sick | positive) = 9 ÷ 98 ≈ **9%**
- Raise "how common" to 10%: the same positive test means ≈ **53%**



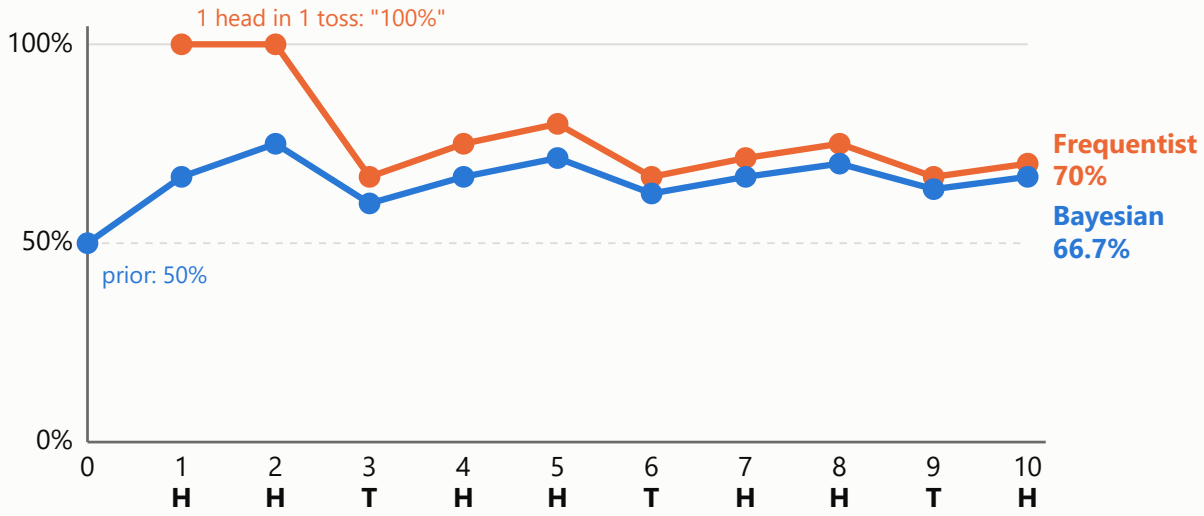
98 positives, only 9 sick → P(sick | positive) = 9 ÷ 98 ≈ 9%

KEY INSIGHT

When the cause is rare, most positive results are false alarms, even with a good test.

Frequentist vs Bayesian: 7 heads in 10 tosses

- **Frequentist:** "The true probability is a fixed but unknown number. Let's estimate it from the data." → 7 ÷ 10 = **70%**
- **Bayesian:** "I start with some uncertainty and combine that prior belief with the evidence." With no preference (a uniform prior, 50%): **`(heads + 1) ÷ (tosses + 2)`** = 8 ÷ 12 = **66.7%**, updated after every toss



KEY INSIGHT

Frequentist: estimate the probability from the data. Bayesian: start from a prior and update it with every toss.

With more data, the two agree

Question	Frequentist	Bayesian
Estimate after 10 tosses (7 heads)	7 ÷ 10 = **70%**	8 ÷ 12 = **66.7%**
Uses a prior?	no	yes: here, no preference (uniform)
How is uncertainty expressed?	for example, a confidence interval	the whole posterior distribution, for example a credible interval
After 1,000 tosses (700 heads)	700 ÷ 1,000 = **70%**	701 ÷ 1,002 ≈ **70.0%**

KEY INSIGHT

With little data the prior matters; with lots of data the evidence decides, and the two answers meet.

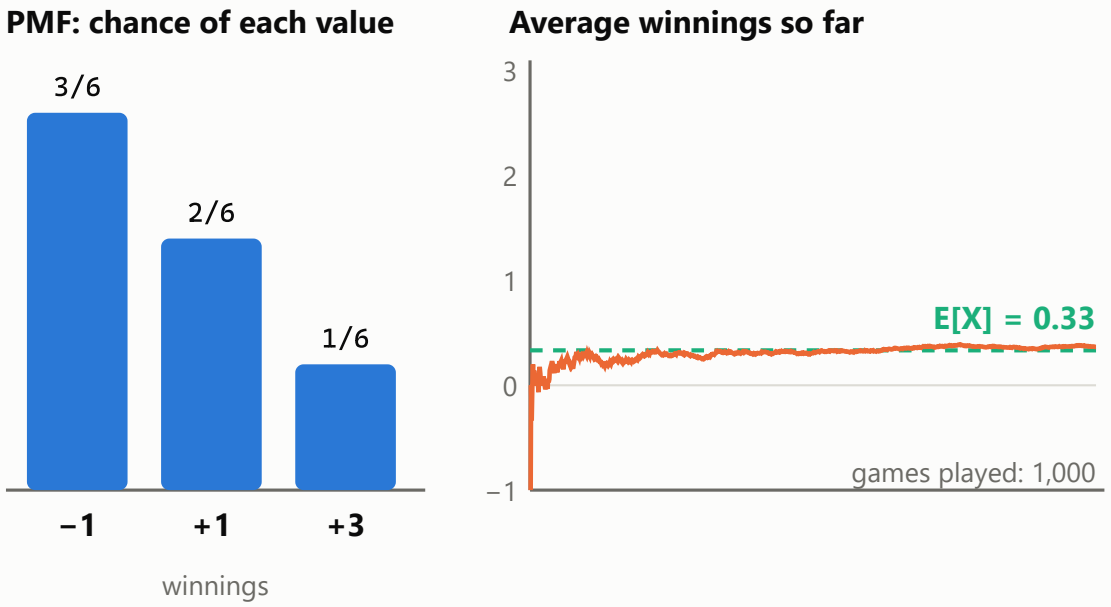
PART C

Random variables

7 Random variables
8 Common distributions
9 More than one variable

The expected value is the long-run average

- A **random variable** X is a number that depends on chance; its **distribution** lists its values and their chances
- Dice game: lose 1 on 1-3 (3/6), win 1 on 4-5 (2/6), win 3 on 6 (1/6)
- **`E[X] = Σ value × probability`** = −1 × 3/6 + 1 × 2/6 + 3 × 1/6 ≈ **0.33** points per game
- Spread: Var(X) ≈ 2.22, standard deviation σ ≈ **1.49** points

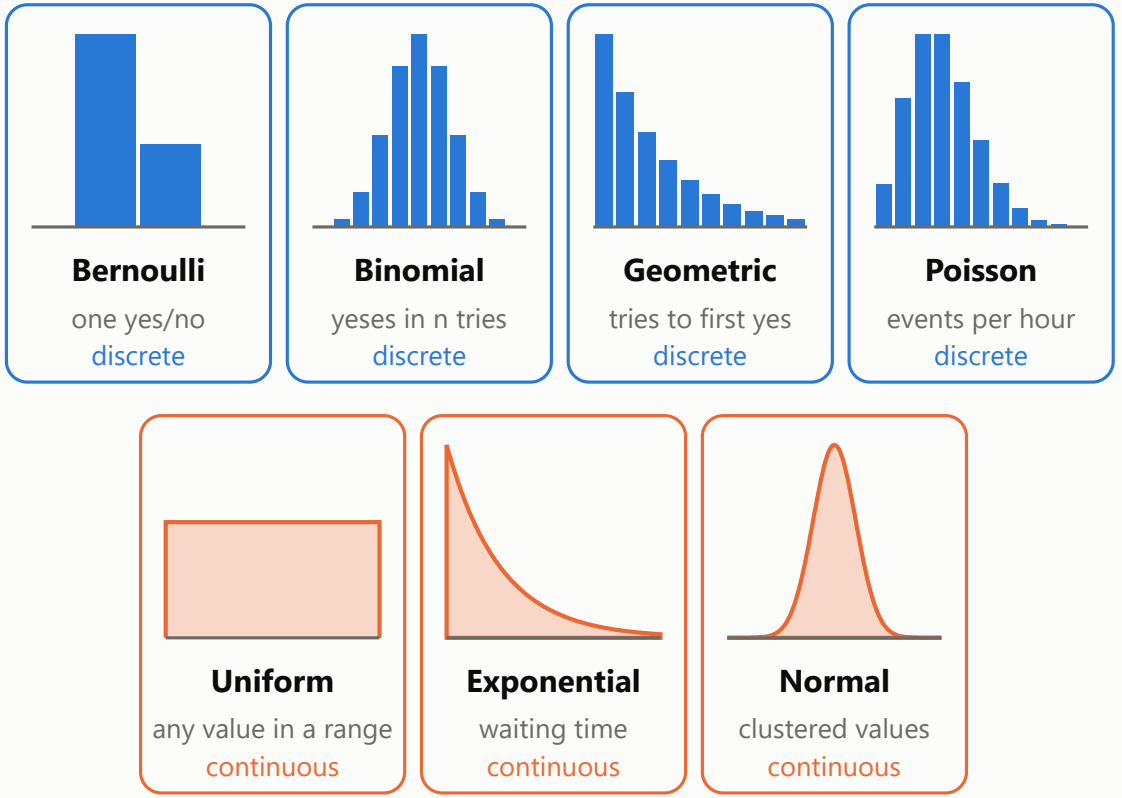


KEY INSIGHT

You never win exactly 0.33 points in one game; it's what the average settles to over many games.

Recognise the pattern, pick the distribution

- A **distribution** is a ready-made pattern for chance; its **parameters** set the details
- Count yeses in n tries → **Binomial** (average n × p); events per hour → **Poisson** (average λ); waiting time → **Exponential**; clustered measurements → **Normal** (average μ, spread σ)
- Also: Bernoulli (one yes/no), Geometric (tries until the first yes), Uniform (any value in a range)

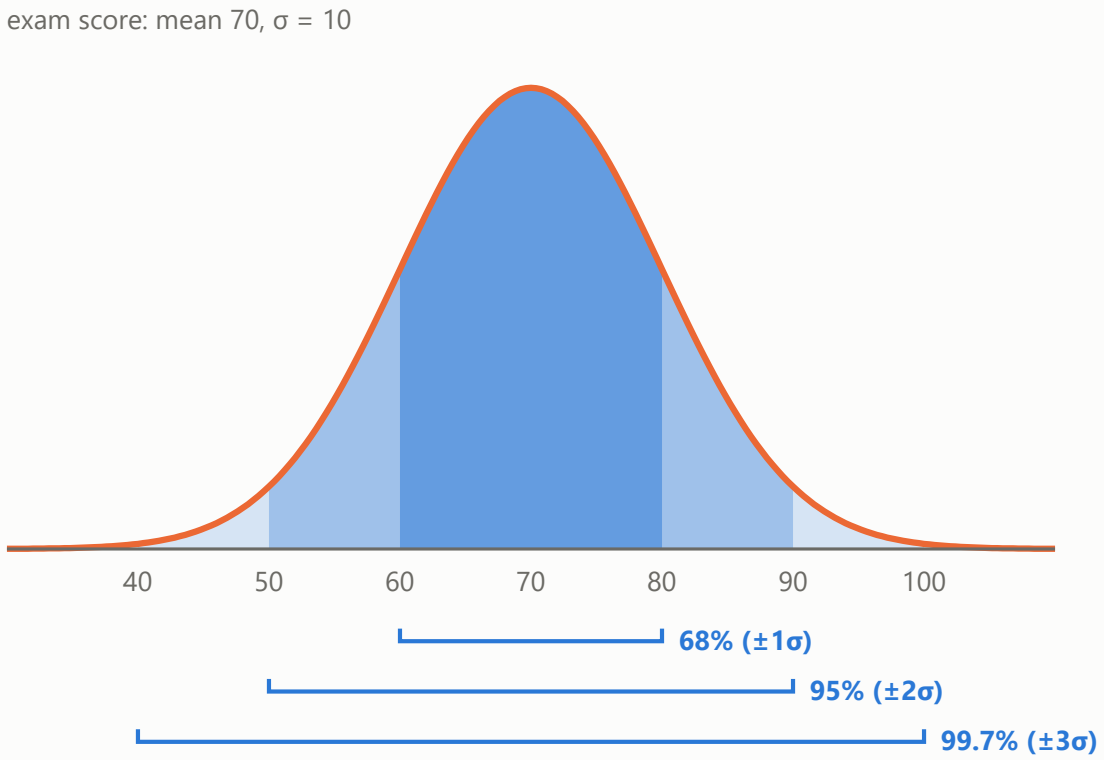


KEY INSIGHT

Recognise the situation, pick the matching distribution, set its parameter: you get its probabilities, average and spread for free.

The 68-95-99.7 rule

- For a **Normal** distribution with mean μ and standard deviation σ: about **68%** of values fall within 1σ of the mean, **95%** within 2σ, and **99.7%** within 3σ
- Exam scores averaging 70 with σ = 10: about 95% score between **50 and 90**

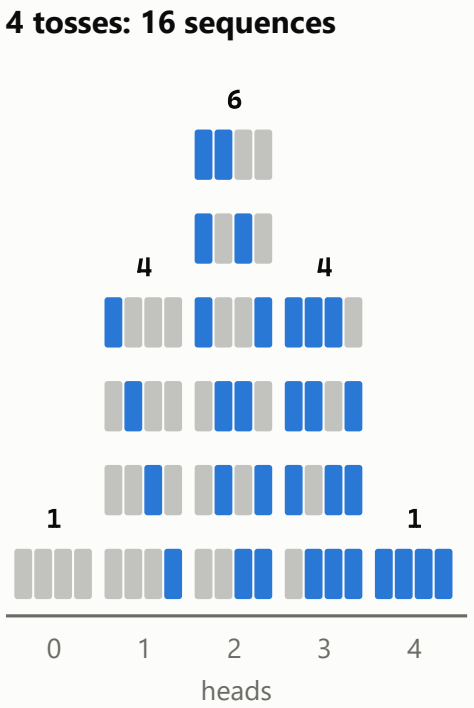


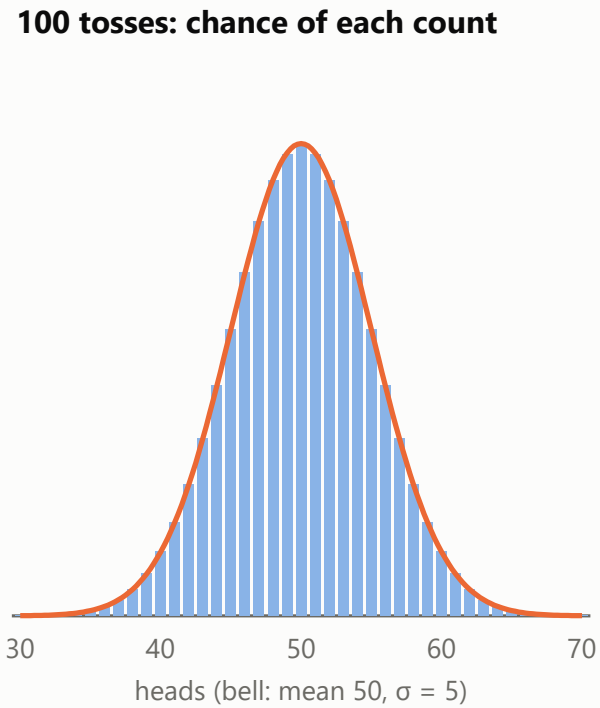
KEY INSIGHT

For bell-shaped data, the mean and the standard deviation tell you where almost everything lies.

Add up many small effects and you get a bell

- One coin toss is just 0 or 1. Toss 100 coins, count the heads, and repeat: the counts pile up in a bell around 50
- **Why the middle wins:** far more sequences land near the middle. 4 tosses: ways to get 0-4 heads = 1, 4, 6, 4, 1
- **Central Limit Theorem:** the sum or average of many independent values tends to be Normal, whatever each single value looks like





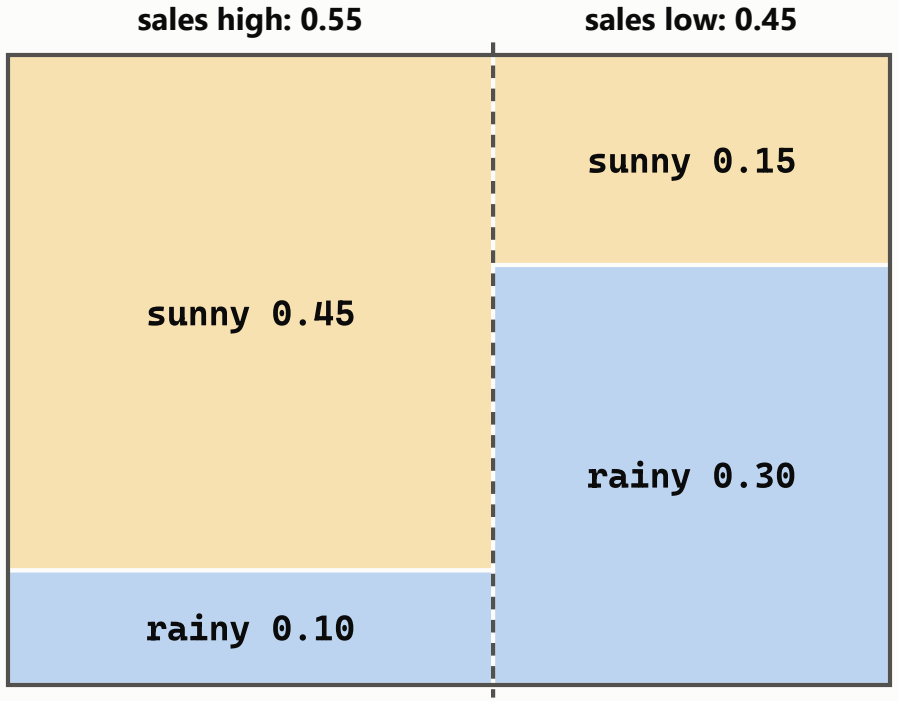
KEY INSIGHT

Heights, errors and exam scores are bell-shaped because each is a sum of many small, independent effects.

Two rules hold everything together

	Sales high	Sales low	Total
Sunny	0.45	0.15	0.60
Rainy	0.10	0.30	0.40
Total	0.55	0.45	1

- **Sum rule:** P(sunny) = 0.45 + 0.15 = 0.60
- **Product rule:** P(sunny, high) = P(sunny) × P(high | sunny) = 0.60 × 0.75 = 0.45
- Not independent: 0.60 × 0.55 = 0.33 ≠ 0.45



sum rule: sunny = 0.45 + 0.15 = 0.60

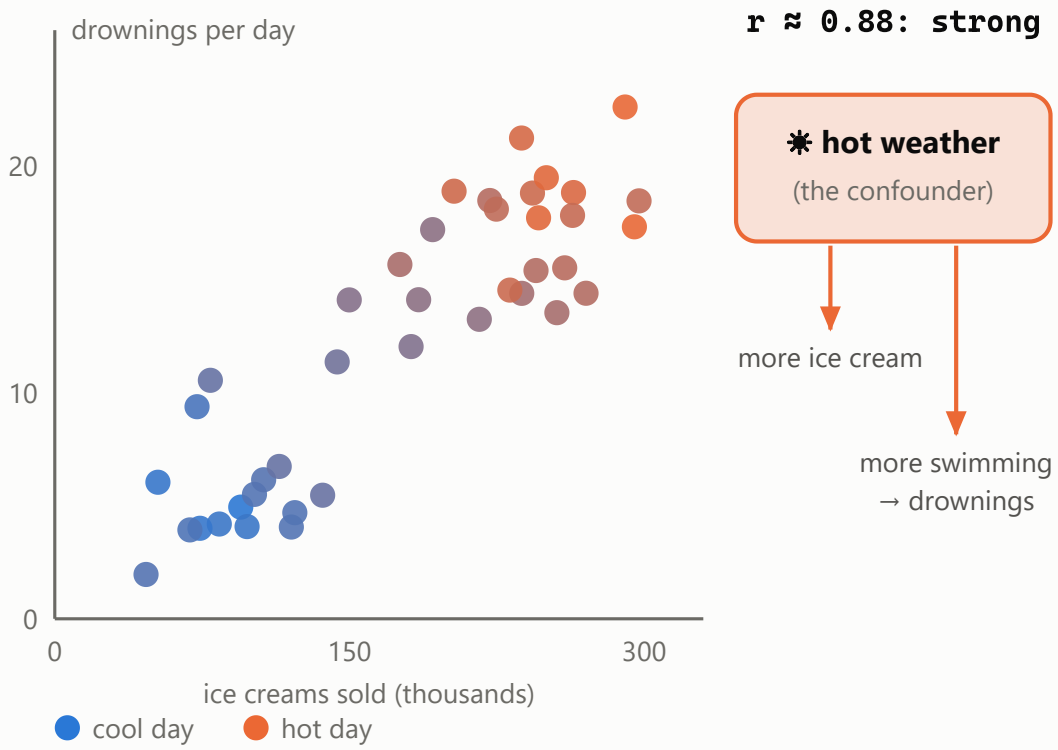
area = probability; the whole square = 1

KEY INSIGHT

The sum rule and the product rule: conditional probability, total probability and Bayes all follow from these two.

Correlation is not causation

- **Covariance** gives the direction: do two numbers tend to be above their averages at the same time?
- **Correlation r** gives direction and strength without units: −1 (perfect opposite) through 0 to +1 (perfect together)
- Ice cream sales and drownings have a strong positive correlation, but hot weather drives both: a **confounder**



KEY INSIGHT

Correlation tells you that two things move together, not why.

Key takeaways

1. Probability measures how likely, from 0 to 1; many trials reveal it.
2. Start every problem with Ω and the event as a set.
3. Choices multiply; divide by k! when order doesn't matter.
4. Overlaps: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
5. "At least one" = 1 − P(none).
6. Conditioning zooms in, and $P(A \mid B) \neq P(B \mid A)$.
7. Bayes: always ask for the base rate; a Bayesian updates a prior with every new piece of evidence.
8. The expected value is the long-run average.
9. Sums of many small effects make a bell (CLT).
10. Correlation is not causation.

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KEY INSIGHT

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